

Appendix: Derivation of Variance scaling factor through second moment of the centred adjacency matrix B

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We have successfully centered the adjacency matrix so that each off-diagonal entry of B is drawn from a distribution with a mean of 0. Now, we must introduce a scaling factor so that the asymptotic variance matches the requirements of a Wigner matrix.

Consider the second empirical spectral moment of B :

$$\frac{1}{n} \text{Tr}(B^2) = \frac{1}{n} \sum_{i=1}^n (B^2)_{ii} \quad (1)$$

By the definition of matrix multiplication, the diagonal entries are given by:

$$(B^2)_{ii} = \sum_{j=1}^n B_{ij} B_{ji} = \sum_{j=1}^n B_{ij}^2 \quad (2)$$

since B is symmetric ($B_{ij} = B_{ji}$). Because the graph has no self-loops, the diagonal entries of the adjacency matrix are zero ($B_{ii} = 0$), allowing us to isolate the off-diagonal entries:

$$\frac{1}{n} \text{Tr}(B^2) = \frac{1}{n} \sum_{i \neq j} B_{ij}^2 \quad (3)$$

Taking the expectation to find the second theoretical spectral moment, we apply the linearity of expectation:

$$\frac{1}{n} \mathbb{E}[\text{Tr}(B^2)] = \frac{1}{n} \sum_{i \neq j} \mathbb{E}[B_{ij}^2] \quad (4)$$

Since $\mathbb{E}[B_{ij}] = 0$ for all $i \neq j$, the expected squared entry is exactly equal to the variance: $\mathbb{E}[B_{ij}^2] = \text{Var}[B_{ij}] = p(1-p)$. Summing over all $n(n-1)$ off-diagonal pairs yields:

$$\begin{aligned} \frac{1}{n} \mathbb{E}[\text{Tr}(B^2)] &= \frac{1}{n} \cdot n(n-1)p(1-p) \\ &= (n-1)p(1-p) \end{aligned} \quad (5)$$

As $n \rightarrow \infty$, this theoretical second moment increases without bound. To restrict the spectral domain to a compact interval, we normalize the matrix by defining:

$$H = \frac{B}{\sqrt{np(1-p)}} \implies H^2 = \frac{B^2}{np(1-p)} \quad (6)$$

Calculating the expected second moment of our modified matrix H yields:

$$\mathbb{E}[m_2^{(n)}] = \frac{1}{n} \mathbb{E}[\text{Tr}(H^2)] = \frac{(n-1)p(1-p)}{np(1-p)} = \frac{n-1}{n} \quad (7)$$

Taking the limit as the graph dimension expands to infinity:

$$\lim_{n \rightarrow \infty} \frac{n-1}{n} = 1 \quad (8)$$

Asymptotically as $n \rightarrow \infty$, the second expected spectral moment of the centered and scaled adjacency matrix converges precisely to 1, which perfectly matches the second moment ($C_1 = 1$) of Wigner's theoretical semicircle distribution.